

Question 16**1 mark**

Identify the number of x -intercepts on the graph of the polynomial function,

$$p(x) = 3x(x - 8)(x^2 + 5).$$

- a. 1
- b. 2
- c. 3
- d. 4

Question 17**1 mark**

Identify the equation that represents the graph of $y = x^2 - 3$ after a reflection over the x -axis.

- a. $y = -x^2 - 3$
- b. $y = x^2 - 3$
- c. $y = x^2 + 3$
- d. $y = -x^2 + 3$

Question 18**1 mark**

Given $2^{\log_2 3} = x$, identify the value of x .

- a. 1
- b. 3
- c. 6
- d. 8

Question 19**1 mark**

Identify the y-intercept on the graph of $y = \frac{x-3}{x^2-5x+6}$.

- a. -3
- b. -2
- c. $-\frac{1}{2}$
- d. 3

Question 20**1 mark**

Identify the total number of arrangements for 10 students and 2 teachers to sit in a row if the teachers must sit together.

- a. $10!2!$
- b. $11!$
- c. $11!2!$
- d. $12!$

Question 21**1 mark**

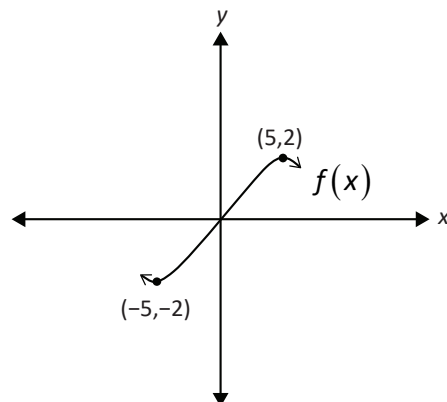
Identify which expression represents the remainder when the polynomial $P(x)$ is divided by $(x-4)$.

- a. $P(-4)$
- b. $P(4)$
- c. $P(x-4)$
- d. $P(x+4)$

Question 22**1 mark**

Identify the period of the graph of the sinusoidal function.

- a. 10π
- b. 10
- c. 20π
- d. 20

**Question 23****1 mark**

Identify an equivalent expression for $\log_3 9 + \log_3 5$.

- a. $\log_3 14$
- b. $2\log_3 5$
- c. $\log_3 45$
- d. $\log_3 \left(\frac{9}{5} \right)$

Question 24**1 mark**

Identify the coterminal angle(s) of $\theta = -\frac{11\pi}{10}$ over the interval $[-3\pi, \pi]$.

- a. $\theta = \frac{9\pi}{10}$
- b. $\theta = -\frac{31\pi}{10}, -\frac{21\pi}{10}$
- c. $\theta = -\frac{31\pi}{10}, \frac{9\pi}{10}$
- d. $\theta = -\frac{21\pi}{10}, \frac{9\pi}{10}$

Question 25**3 marks**

117

Express $p(x) = x^3 - 13x - 12$ in completely factored form.

$p(x) =$ _____

Question 26

3 marks

118

Verify, by substitution, that the equation $\frac{\cos^2\theta}{\sin^2\theta} = \frac{2\csc\theta}{\sec^2\theta}$ is true for $\theta = \frac{\pi}{6}$.

Left-Hand Side	Right-Hand Side

Question 27

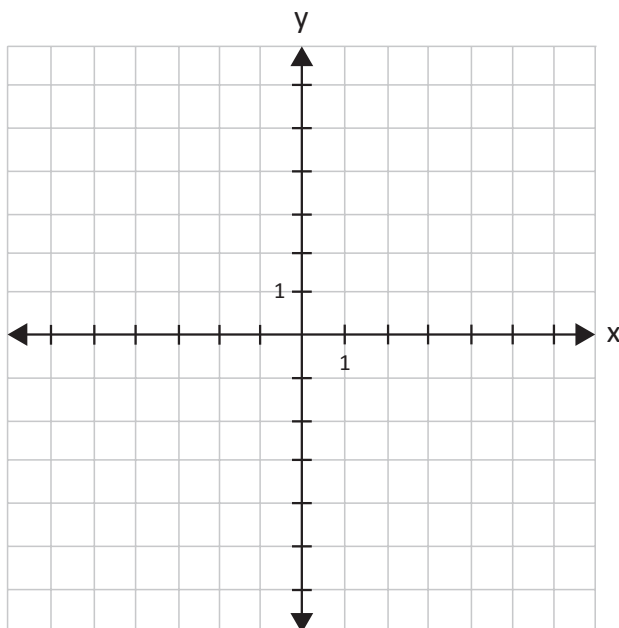
a) 1 mark b) 2 marks

119
120

Given $f(x) = \log_3 x + 2$,

a) determine the x-intercept of $f(x) = \log_3 x + 2$.

b) sketch the graph of $f(x) = \log_3 x + 2$.



Question 28

2 marks

121

Evaluate.

$$\sec^2\left(\frac{5\pi}{4}\right) \cdot \tan^2\left(-\frac{2\pi}{3}\right)$$

Question 29**1 mark**

122

Given the functions $f(x) = x - 3$ and $g(x) = \sqrt{x} + 2$, state the domain of $g(f(x))$.

Question 30**1 mark**

123

Evaluate.

$$\sin 70^\circ \cos 25^\circ - \cos 70^\circ \sin 25^\circ$$

Question 31

2 marks

124

Solve, algebraically.

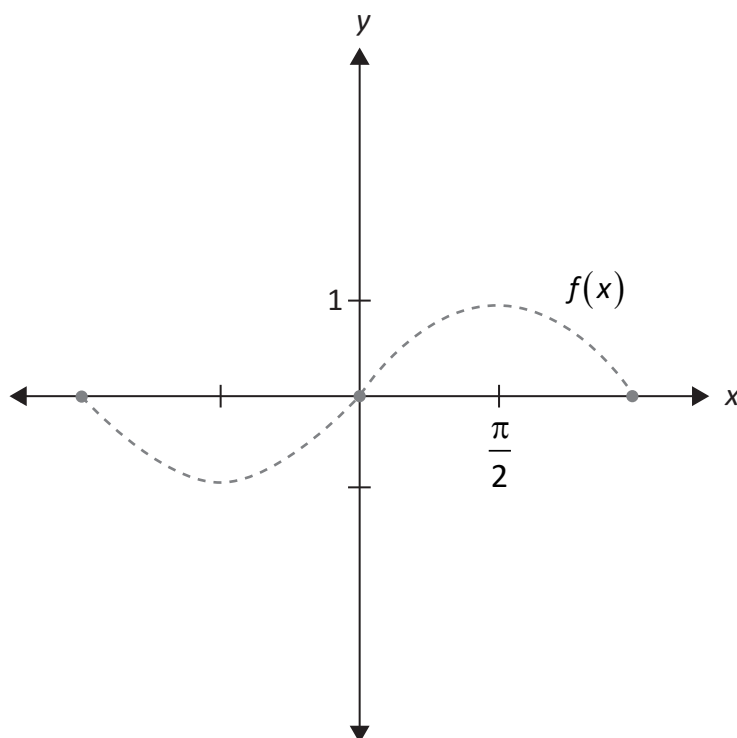
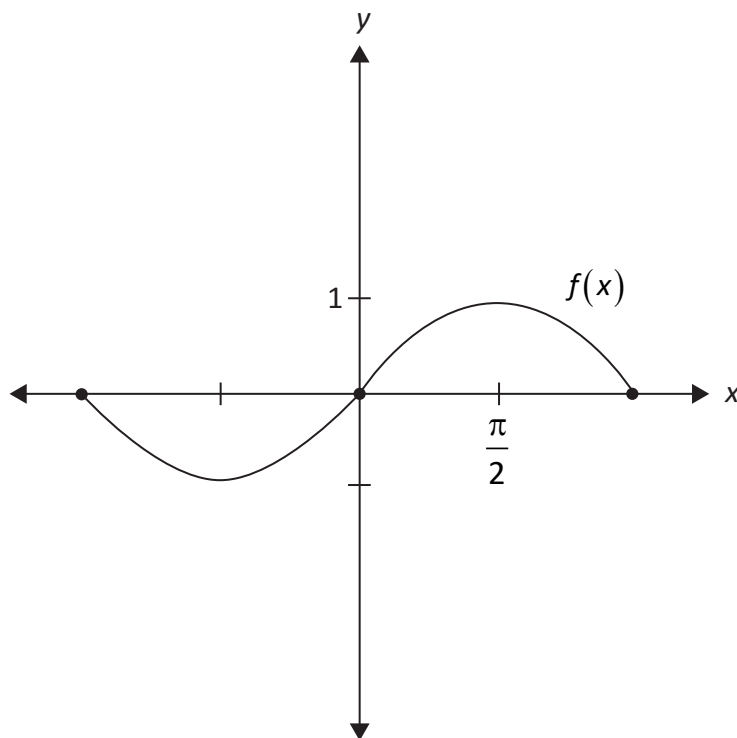
$$\log_{15}(x^2 - 1) = 1$$

Question 32

2 marks

125

Given the graph of $y = f(x)$, sketch the graph of $y = \frac{1}{f(x)}$.



The graph of $f(x)$ has already been drawn for your reference.

No marks will be awarded for the graph of $f(x)$.

Question 33

a) 3 marks b) 1 mark

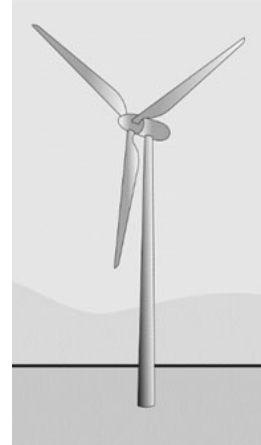
126
127

A light is attached to the end of a blade of a wind turbine. The height of the light, with respect to the ground, follows the sinusoidal equation:

$$h(t) = 80 \cos\left(\frac{\pi}{2}t\right) + 240$$

where $h(t)$ is the height of the light with respect to the ground, measured in feet and,

t is the time, measured in seconds.



- a) Sketch the graph of the height of the light with respect to the ground, over an interval of 6 seconds.



- b) If the turbine begins to rotate faster, describe the resulting effect on the period of the graph of $h(t)$.

Question 34

1 mark

128

Elodie was asked to determine the total number of arrangements of the letters in the word EXCELLENCE.

Her solution:

$$10! = 3628\ 800$$

Describe her error.

Question 35

a) 3 marks b) 1 mark

129
130

Given that $\cos \alpha = -\frac{4}{7}$ where α is in quadrant III and $\sin \beta = \frac{5}{13}$ where β is in quadrant II, determine the exact value of:

a) $\cos(\alpha + \beta)$

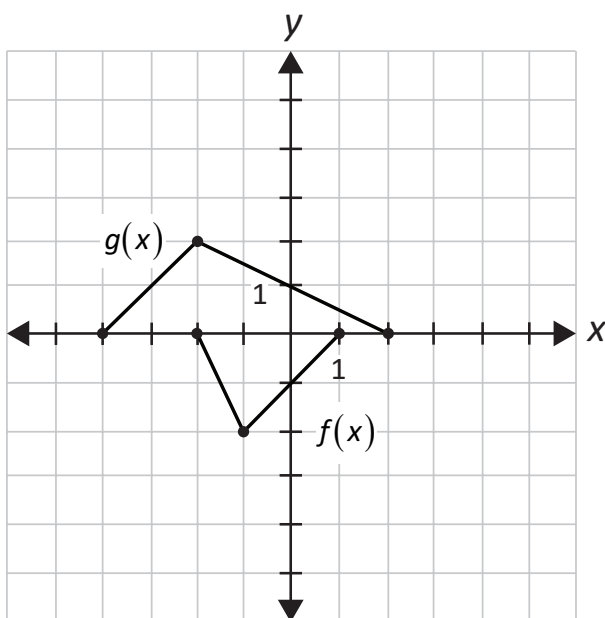
b) $\sec(\alpha + \beta)$

Question 36

2 marks

131

Determine the equation for $g(x)$, in terms of $f(x)$.



$g(x) =$ _____

Question 37

1 mark

132

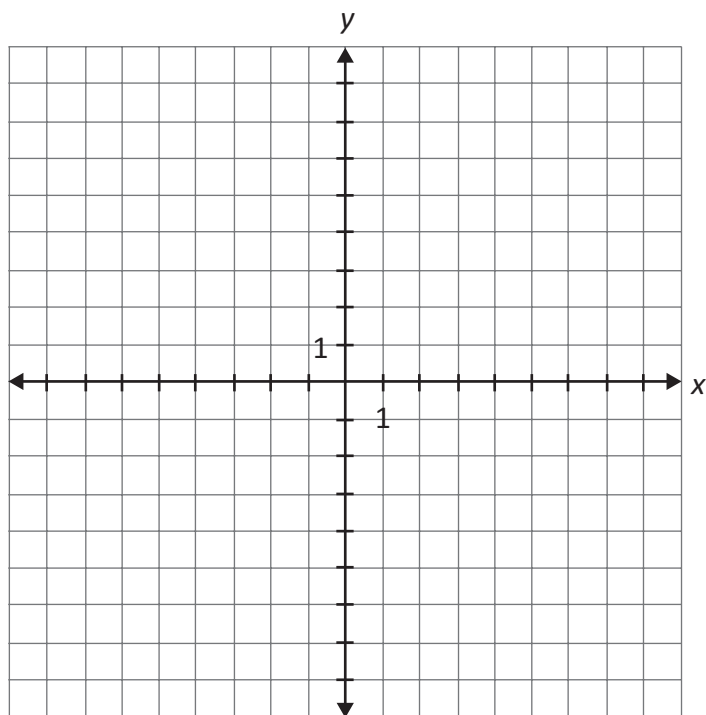
State possible values for m and n that satisfy the equation, $\log_m n = 5$.

Question 38

4 marks

133

Sketch the graph of $f(x) = \frac{x-4}{x^2-7x+12}$.



Question 39

1 mark

134

Justify that the value of ${}_3P_4$ does not exist.

Question 40

1 mark

135

State the range of the sinusoidal function that has the following characteristics:

- an amplitude of 3
- a maximum at $\left(\frac{\pi}{2}, 1\right)$

Question 41

3 marks

136

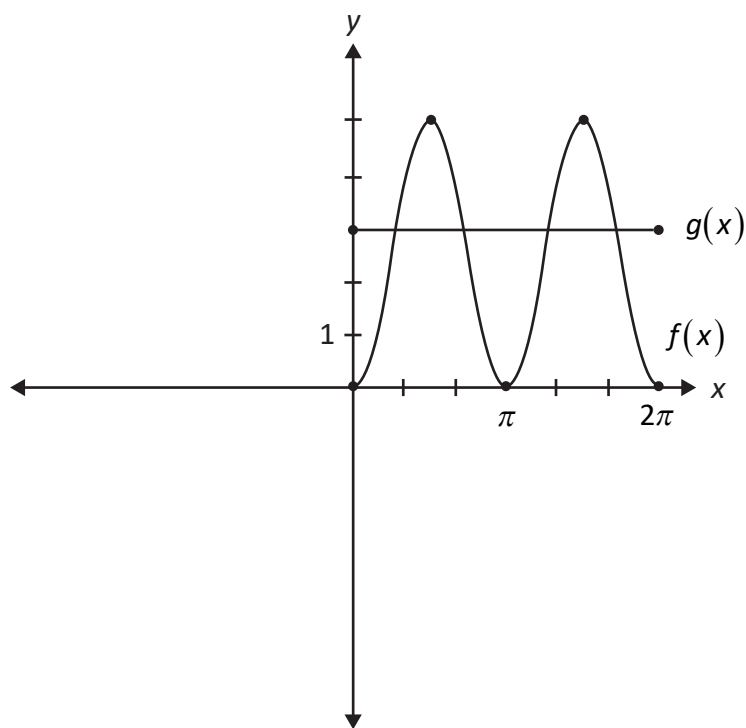
Solve, algebraically.

$${}_nC_2 = 15$$

Question 42**1 mark**

137

Describe how to use the graphs of $f(x) = 5\sin^2 x$ and $g(x) = 3$ to solve the equation, $5\sin^2 x = 3$, over the interval $[0, 2\pi]$.



Question 43**3 marks**

138

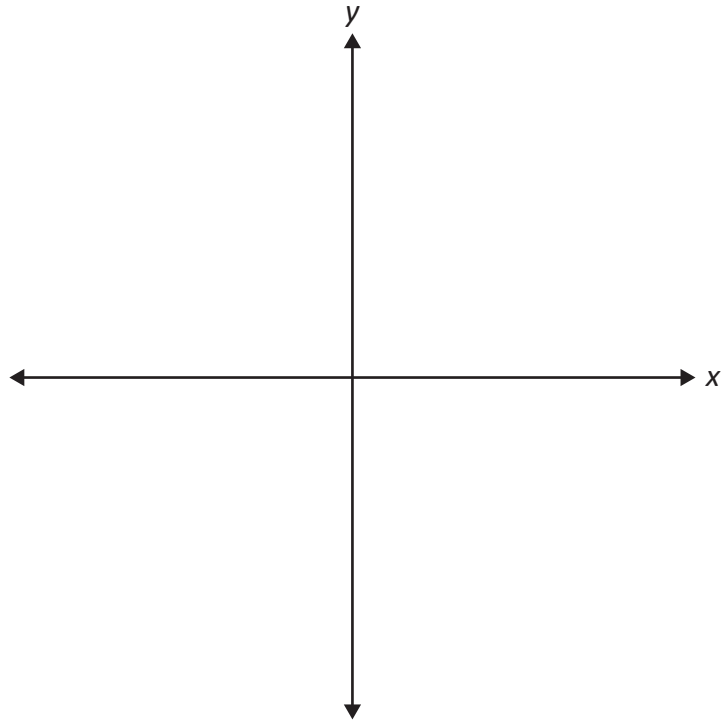
Solve, algebraically.

$$2\log_3 5 - \frac{1}{3}\log_3 125 = \log_3 a$$

Question 44**1 mark**

139

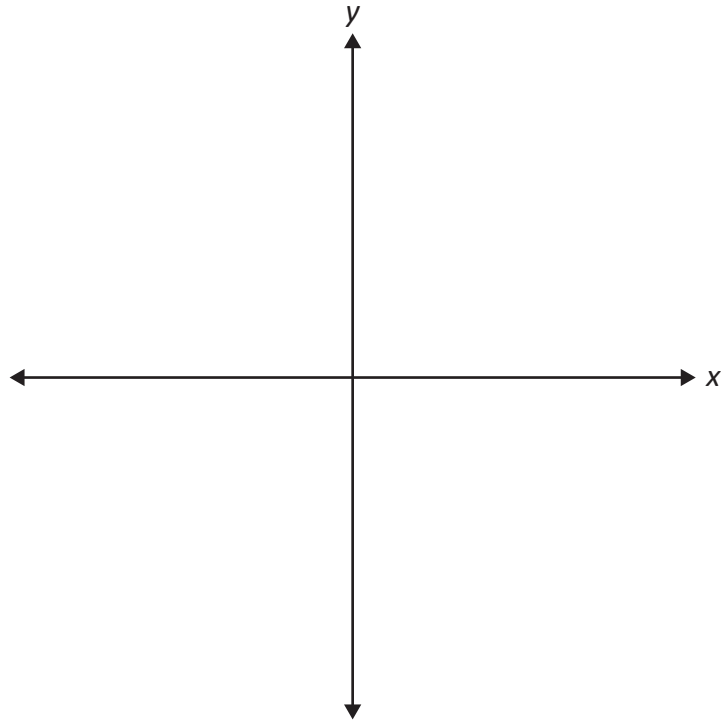
Sketch the angle, $\frac{-7\pi}{12}$, in standard position.



Question 45**3 marks**

140

Sketch the graph of $p(x) = -(x-1)(x-2)(x+4)^2$.



Question 46**2 marks**

141

The point, $P(\theta) = \left(-\frac{1}{6}, y\right)$, lies on the unit circle and is located in Quadrant III. Determine the exact value of $\csc \theta$.

NO MARKS WILL BE AWARDED FOR WORK DONE ON THIS PAGE.